MATHEMATICS FOR ENGINEERING



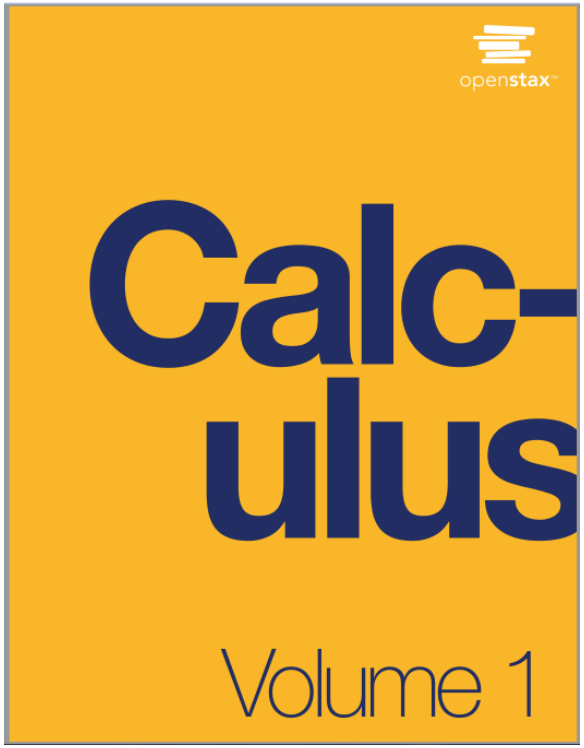
MAE101

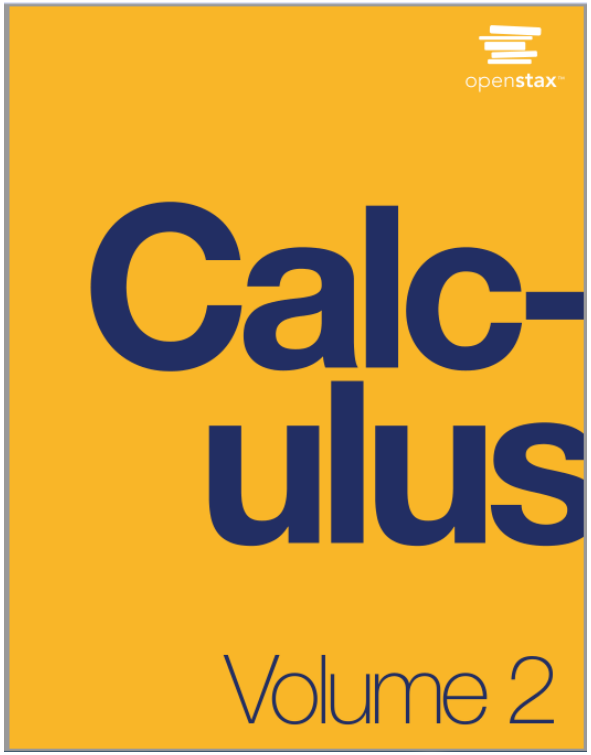
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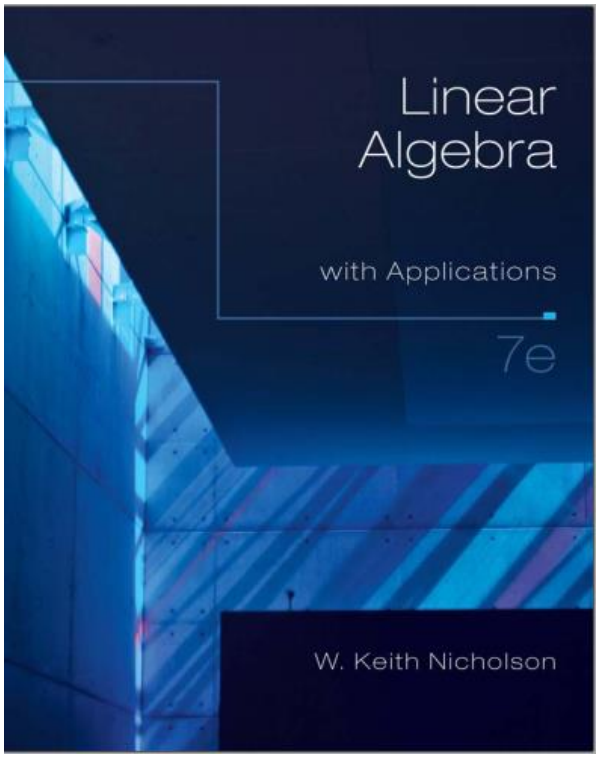
Evaluation Strategy

- Must attend more than 80% of contact hours
- Evaluating
  - 03 Progress tests: → 30%
  - 03 Assignments: → 30%
  - Final Exam (FE): → 40%
- Pass: Every on-going assessment component > 0
  and Final Exam score ≥ 4 (of 10)
  and Final Result ≥ 5
  Retake only the Final Exam when not passed

Materials


openstax
Calc-
ulus
Volume 1


openstax
Calc-
ulus
Volume 2


Linear
Algebra
with Applications
7e
W. Keith Nicholson

Course description

Chapter 1: Functions and Graphs
Chapter 2: Limits
Chapter 3: Derivatives
Chapter 4: Applications of Derivatives
Chapter 5: Integration
Chapter 6: Techniques of Integrations
Chapter 1: System of Linear Equations
Chapter 2: Matrix Algebra
Chapter 3: Determinants and Diagonalization
Chapter 4: Vector geometry
Chapter 5: The vector space R^n

Chapter 1: Function and Graphs

(Page 5-94, Calculus Volume 1)

Objectives

- Four ways to represent a function
- Basis functions and the transformations of functions

CONTENTS

1.1 Review of Functions

1.2 Basic Classes of Functions

1.3 Trigonometric Functions

1.4 Inverse Functions

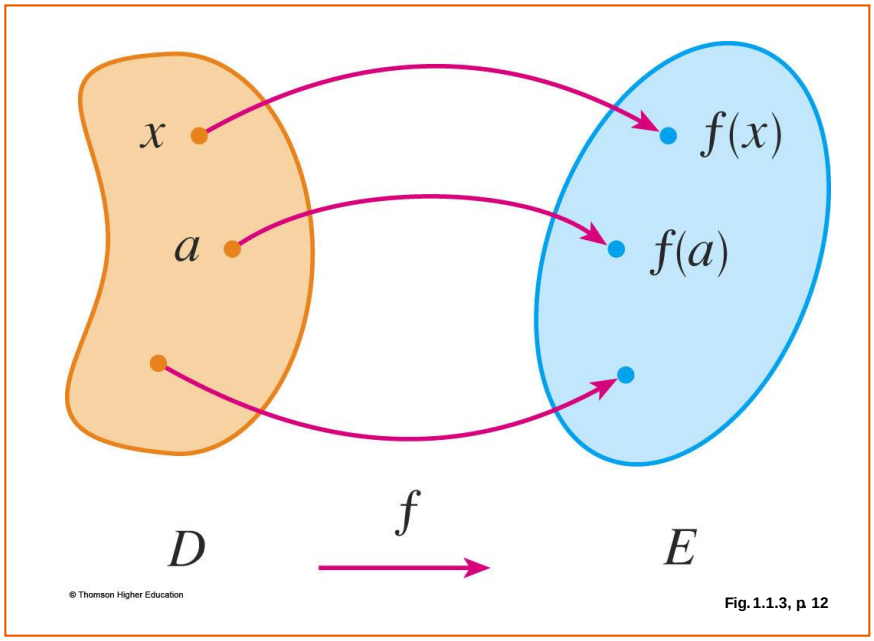
1.5 Exponential and Logarithmic Functions

FUNCTION

A **function** *f* is a rule that assigns to each element *x* in a set ***D*** *exactly one* element, called *f*(*x*), in a set ***E***.

The set *D* is called the **domain** of the function *f*.

The **range of *f*** is the set of all possible values of *f*(*x*) as *x* varies throughout the domain.

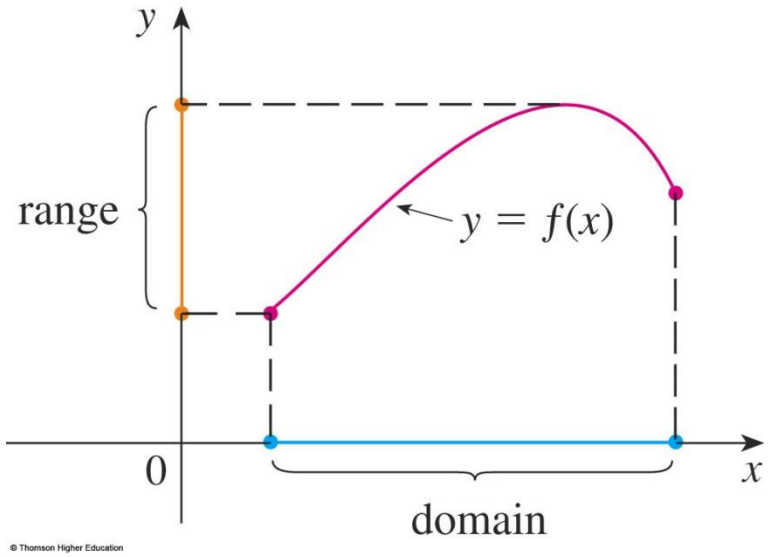


GRAPH

The **graph** of *f* is the **set** *of all points (x, y) in the coordinate plane such that y = f(x) and x is in the domain of f*.

The graph of *f* also allows us to picture:

- The **domain of *f*** on the *x*-axis
- **Its range** on the *y*-axis



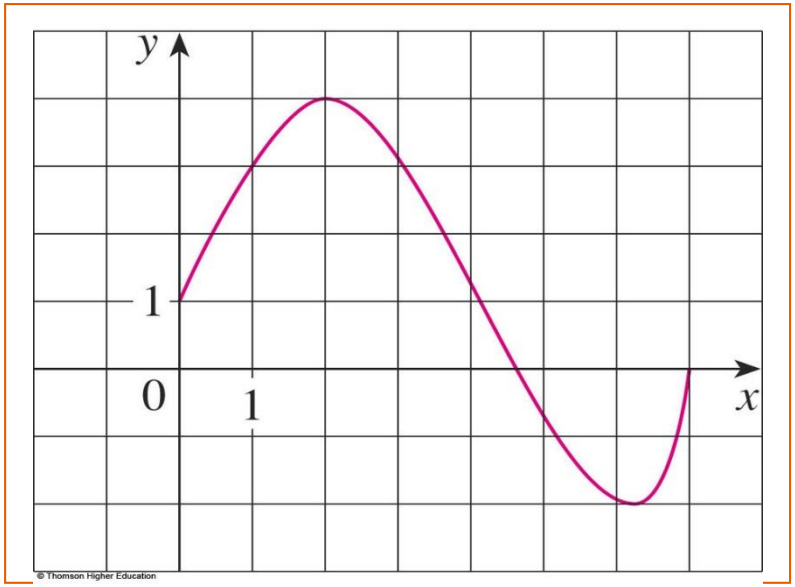
GRAPH

Example 1

The graph of a function f is shown.

a. Find the values of $f(1)$ and $f(5)$.

b. What is the domain and range of f ?



Example

Find the domain and region of the functions (if it is a function)

a. $f(x) = \sqrt{x+2}$

b. $g(x) = \dfrac{1}{x^2 - x}$

REPRESENTATIONS

DISCUSSION

Find the domain and region of the functions (if it is a function)

a. $f(n) = \sqrt{n}$ for all natural numbers n.

b. $g(x)$ is any real number such that larger than x

REPRESENTATIONS OF FUNCTIONS

There are four possible ways to represent a function:

- Algebraically (by an explicit formula)
- Visually (by a graph)
- Numerically (by a table of values)
- Verbally (by a description in words)

EXAMPLE

The human population of the world P depends on the time t.

- The table gives estimates of the world population $P(t)$ at time t, for certain years.
- However, for each value of the time t, there is a corresponding value of P, and we say that P is a function of t.

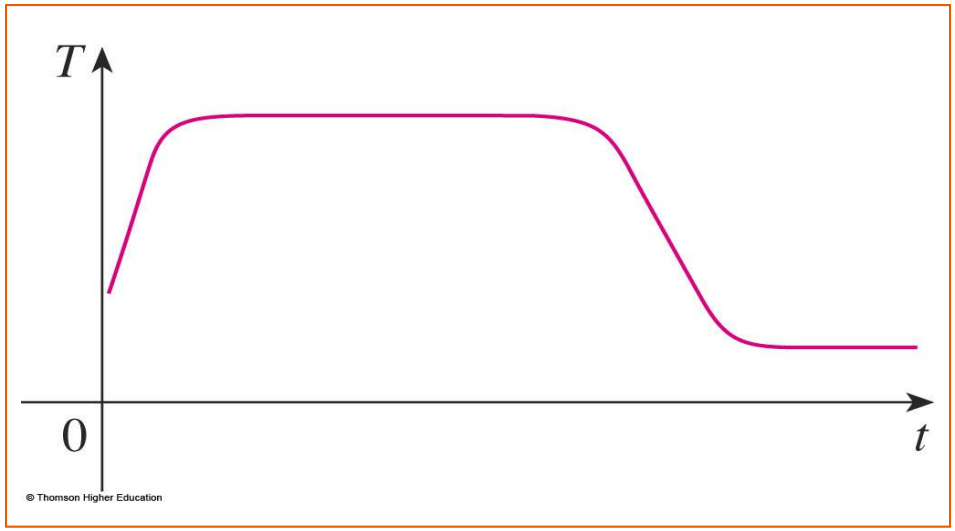
Year	Population (millions)
1900	1650
1910	1750
1920	1860
1930	2070
1940	2300
1950	2560
1960	3040
1970	3710
1980	4450
1990	5280
2000	6080

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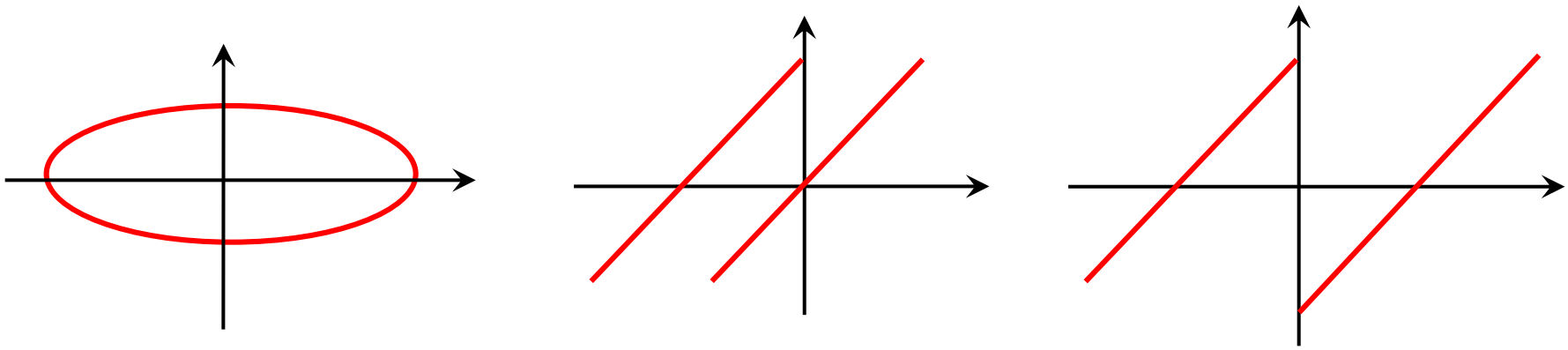
REPRESENTATIONS

"*When you turn on a hot-water faucet, the temperature T of the water depends on how long the water has been running*".

Draw a rough graph of T as a function of the time t that has elapsed since the faucet was turned on.



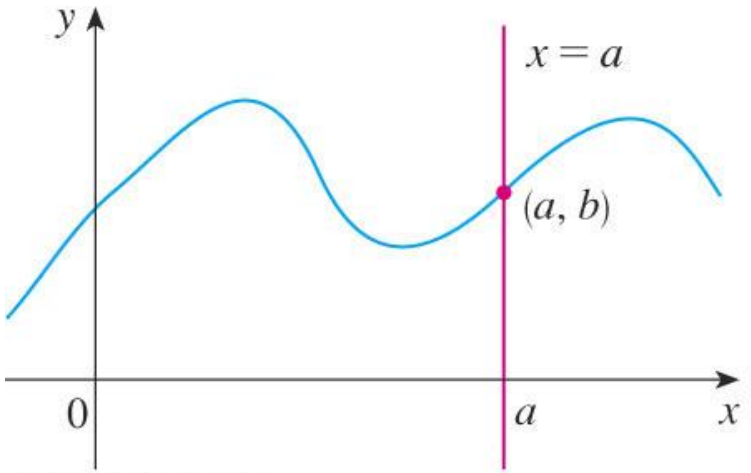
THE VERTICAL LINE TEST



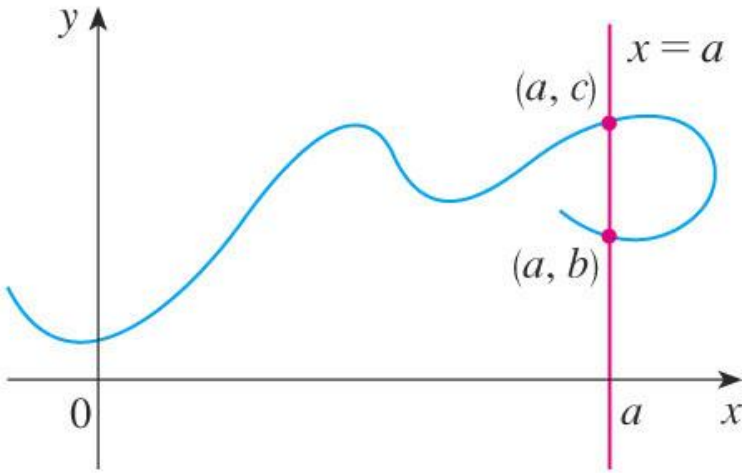
A curve in the *xy*-plane is the graph of a function of *x* if and only if **no vertical line** intersects the curve **more than once**.

THE VERTICAL LINE TEST

The reason for the truth of the Vertical Line Test can be seen in the figure.



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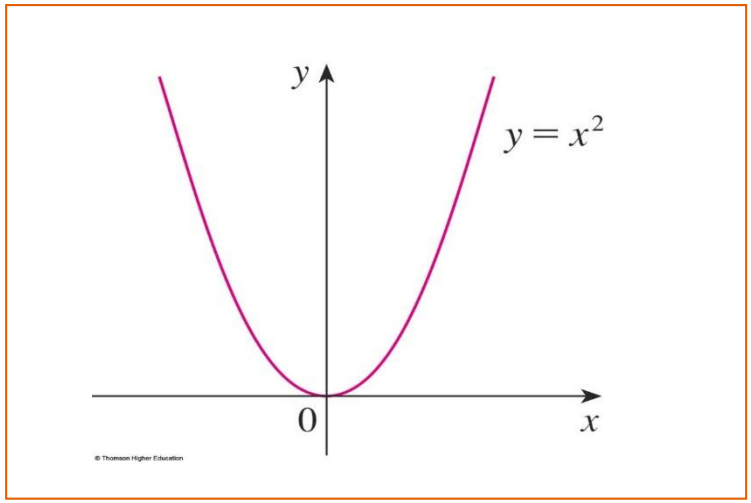
INCREASING AND DECREASING FUNCTIONS

A function f is called **increasing on an interval I** if:

$$f(x_1) < f(x_2) \quad \text{whenever } x_1 < x_2 \text{ in } I$$

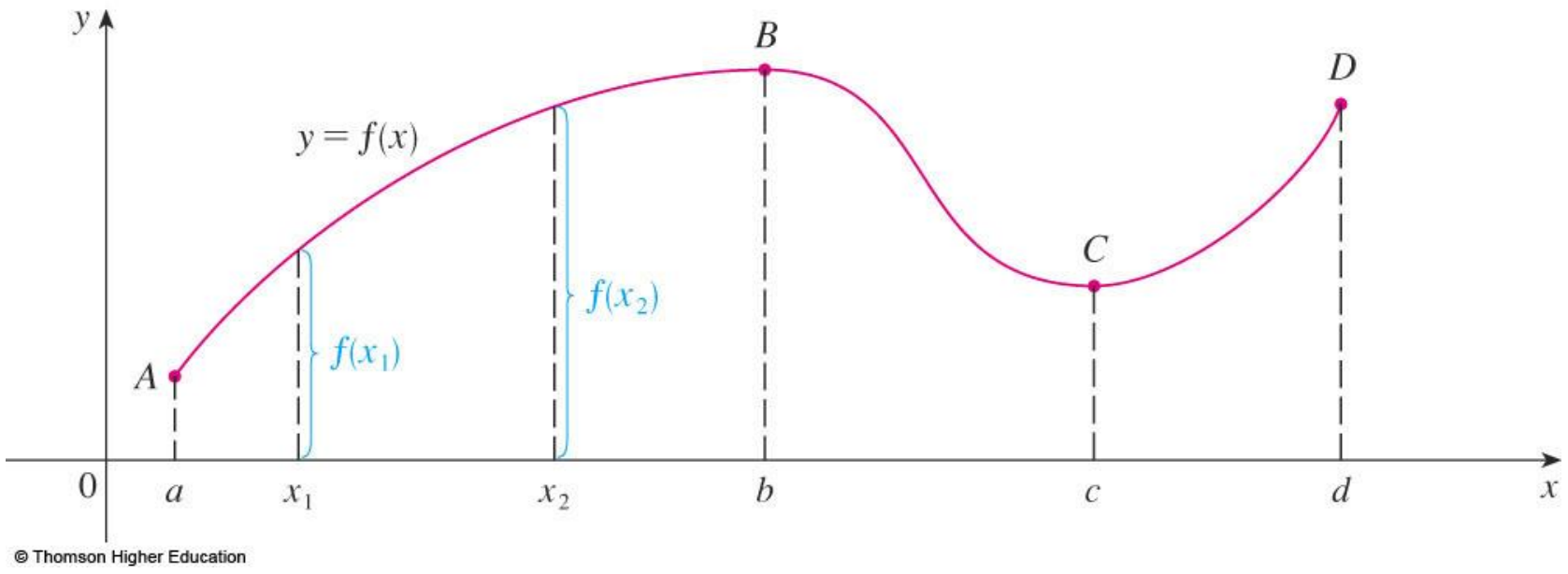
It is called **decreasing on I** if:

$$f(x_1) > f(x_2) \quad \text{whenever } x_1 < x_2 \text{ in } I$$



INCREASING AND DECREASING FUNCTIONS

The function f is said to be **increasing on the interval $[a, b]$**, decreasing on $[b, c]$, and increasing again on $[c, d]$.



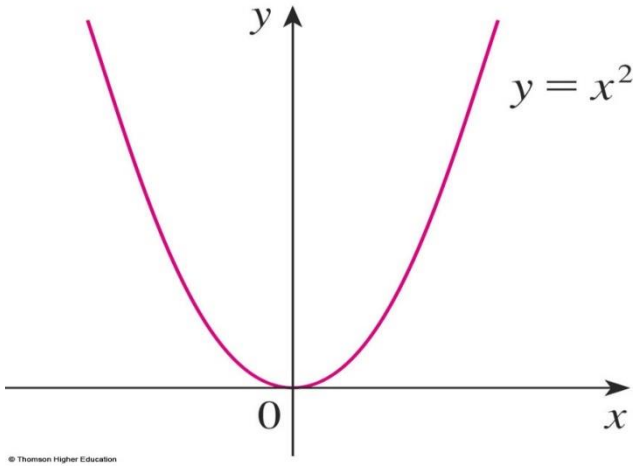
SYMMETRY: EVEN FUNCTION

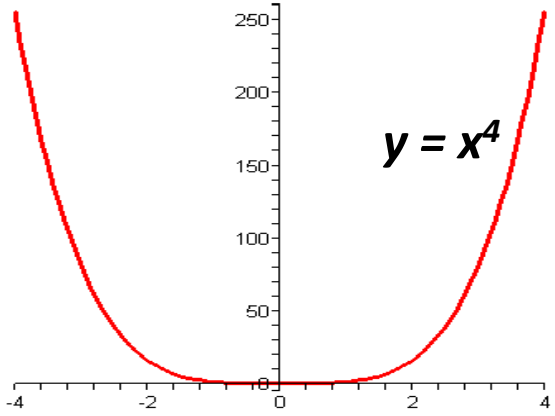
If a function f satisfies:

$$f(-x) = f(x), \text{ for all } x \text{ in } D$$

then f is called an **even function.**

The geometric significance of an even function is that its graph is **symmetric with respect to the y-axis**.





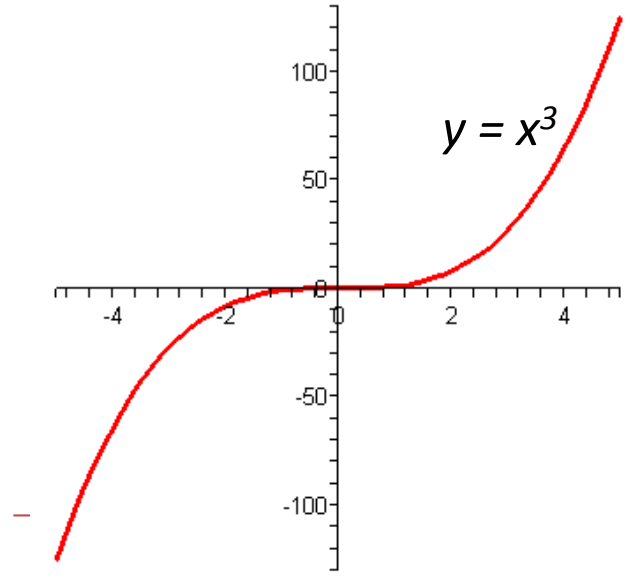
SYMMETRY: ODD FUNCTION

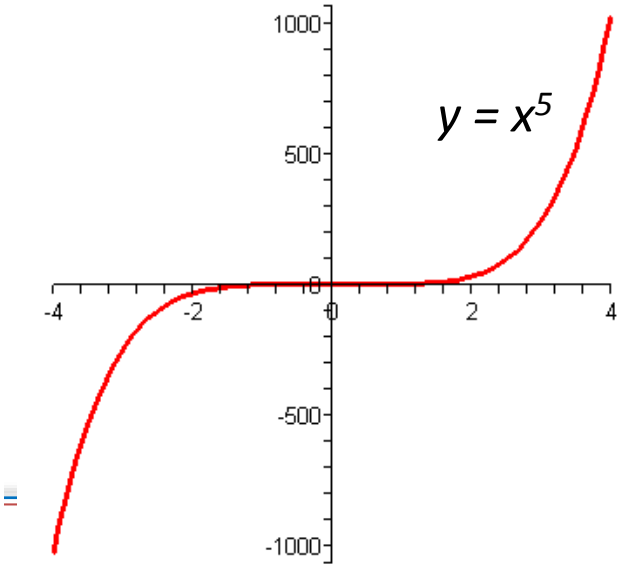
If *f* satisfies:

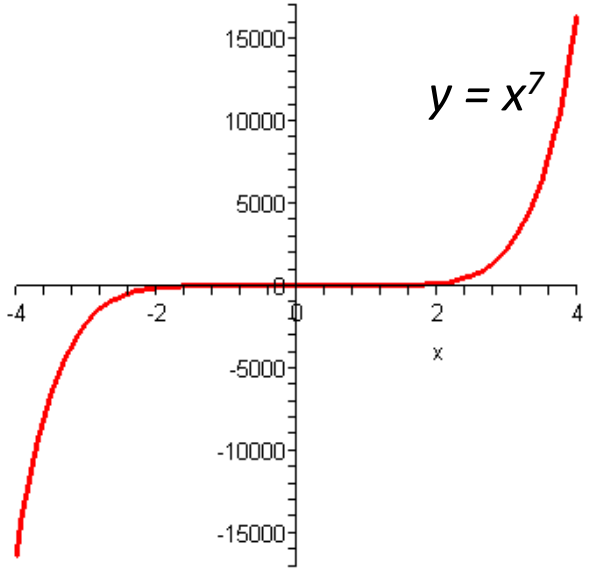
$$f(-x) = -f(x), \text{ for all } x \text{ in } D$$

then *f* is called an **odd function**.

- The graph of an odd function is **symmetric about the origin**.







Example

Let f is an **odd function**. If (-3,5) is in the graph of f then which point is also in the graph of f?

a. (3,5) b. (-3,-5) c. (3,-5) d. All of the others

Example

Suppose f is an odd function and g is an even function.

What can we say about the function f.g defined by (f.g)(x)=f(x)g(x)?

Prove your result.

QUIZ QUESTIONS

1) If f is a function then f(x+2)=f(x)+f(2)

a. True b. False

2) If f(s)=f(t) then s= t

a. True b. False

3) Let f be a function.
We can find s and t such that s=t and f(s) is not equal to f(t)

a. True b. False

COMBINATIONS OF FUNCTIONS

- Two functions *f* and *g* can be combined to form new functions:
  - $(f + g)x = f(x) + g(x)$
  - $(f - g)x = f(x) - g(x)$
  - $(fg)(x) = f(x)g(x)$ $\quad \left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}$
  - $(f \circ g)(x) = f(g(x))$

Let h(x)=f(g(x)).

1) If g(x)=x-1 and h(x)=3x+2 then f(x) is:

a. 3x+3 b. 3x+4 c. 3x+2 d. 3x+5

2) If h(x)=3x+2 and f(x)=x-1 then g(x) is:

a. 3x+2 b. 3x+4 c.3x+3 d. 3x+1

QUIZ QUESTIONS

1) If f and g are functions, then $f \circ g = g \circ f$

a. True b. False

2)

x	1	2	3	4	5	6
$f(x)$	3	2	1	0	1	2
$g(x)$	6	5	2	3	4	6

$(f \circ g)(2)$ is

a. 5 b. 1 c. 2 d. None of the others

1.2
BASIC CLASSES OF FUNCTIONS

ALGEBRAIC FUNCTIONS

LINEAR MODELS

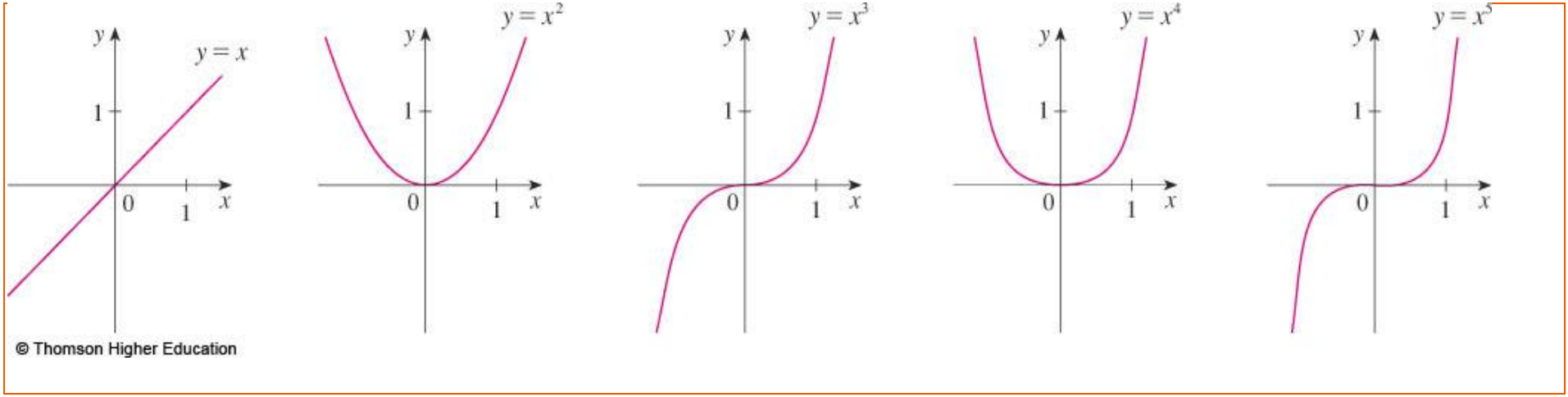
When we say that y is a **linear function** of x, we mean that the graph of the function is a line.

So, we can use the slope-intercept form of the equation of a line to write a formula for the function as $y = f(x) = mx + b$, where m is the slope of the line and b is the y-intercept.

ALGEBRAIC FUNCTIONS

POWER FUNCTIONS

A function of the form $f(x) = x^a$, where a is constant, is called **a power function**.



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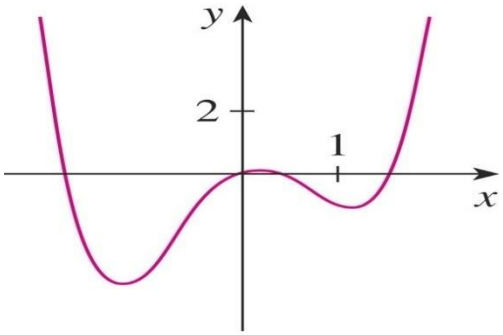
ALGEBRAIC FUNCTIONS

POLYNOMIALS

A function P is called a polynomial if

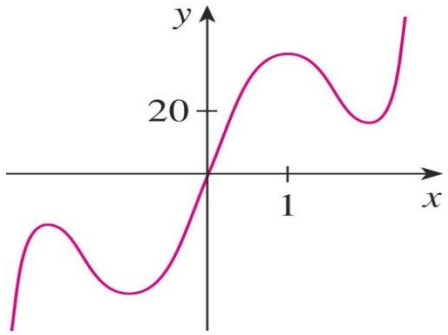
$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \ldots + a_2 x^2 + a_1 x + a_0$$

where a_i are the coefficients of the polynomial.



(b) $y = x^4 - 3x^2 + x$

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(c) $y = 3x^5 - 25x^3 + 60x$

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ALGEBRAIC FUNCTIONS

RATIONAL FUNCTIONS

A rational function *f* is a ratio of two polynomials

$$f(x) = \frac{P(x)}{Q(x)}$$

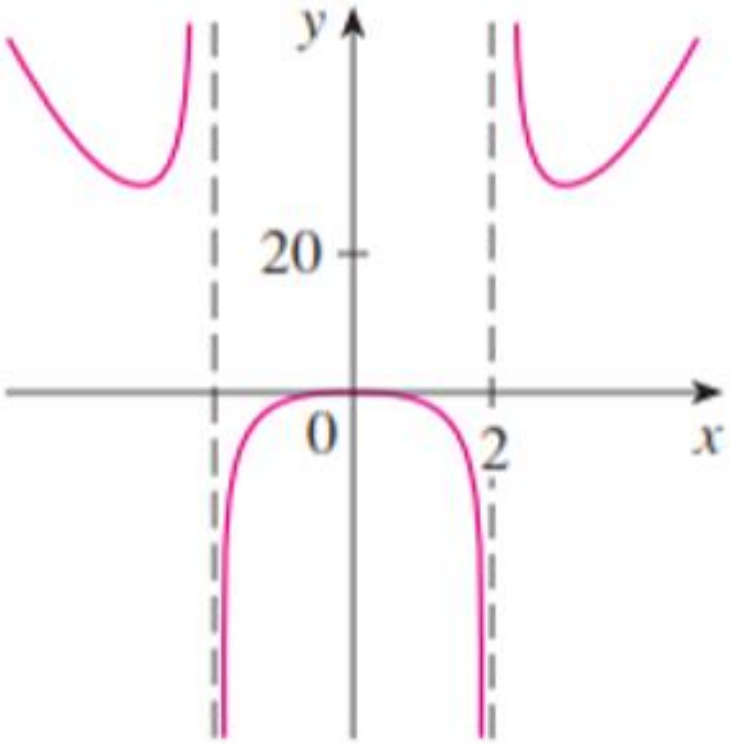
where *P* and *Q* are polynomials. The domain consists of all values of *x* such that

$$Q(x) \neq 0$$

ALGEBRAIC FUNCTIONS

RATIONAL FUNCTIONS

$$f(x) = \frac{2x^4 - x^2 + 1}{x^2 - 4}$$



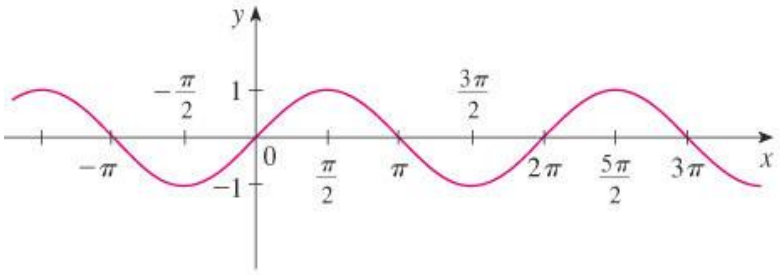
TRANSCENDENTAL FUNCTIONS

TRIGONOMETRIC FUNCTIONS

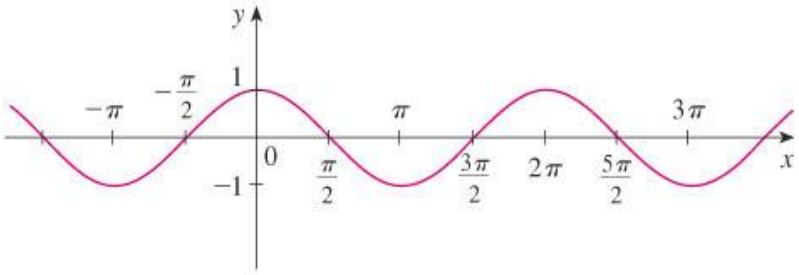
$f(x) = \sin x \qquad D = (-\infty, \infty)$

$g(x) = \cos x \qquad R = [-1, 1]$

$$\sin(x + k2\pi) = \sin x \qquad \cos(x + k2\pi) = \cos x; \quad k \in Z$$



(a) $f(x) = \sin x$



(b) $g(x) = \cos x$

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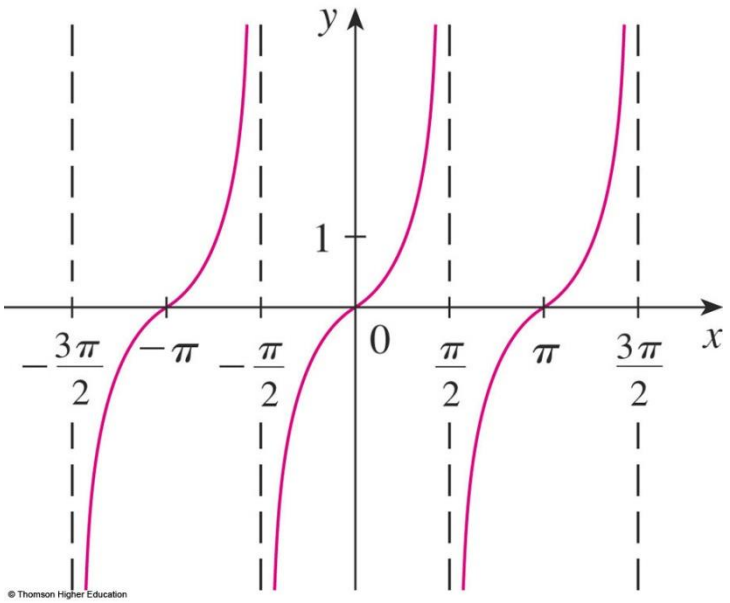
TRANSCENDENTAL FUNCTIONS

TRIGONOMETRIC FUNCTIONS

$$\tan x = \frac{\sin x}{\cos x} \qquad x \neq \pm\frac{\pi}{2}, \pm\frac{3\pi}{2}, \cdots$$

$$R = (-\infty, \infty)$$

$$\tan(x + k\pi) = \tan x; \quad k \in Z$$



TRANSCENDENTAL FUNCTIONS

TRIGONOMETRIC FUNCTIONS

The reciprocals of the sine, cosine, and tangent functions are

$$\csc x = \frac{1}{\sin x}$$

$$\sec x = \frac{1}{\cos x}$$

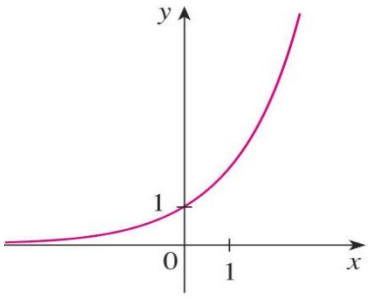
$$\cot x = \frac{1}{\tan x}$$

TRANSCENDENTAL FUNCTIONS

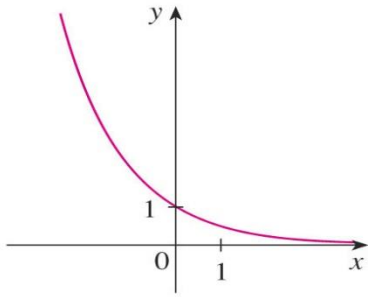
EXPONENTIAL FUNCTIONS

The **exponential functions** are the functions of the form $f(x) = a^x$, where the base a is a positive constant.

- The graphs of $y = 2^x$ and $y = (0.5)^x$ are shown.
- In both cases, the domain is $(-\infty, \infty)$ and the range is $(0, \infty)$.



(a) $y = 2^x$



(b) $y = (0.5)^x$

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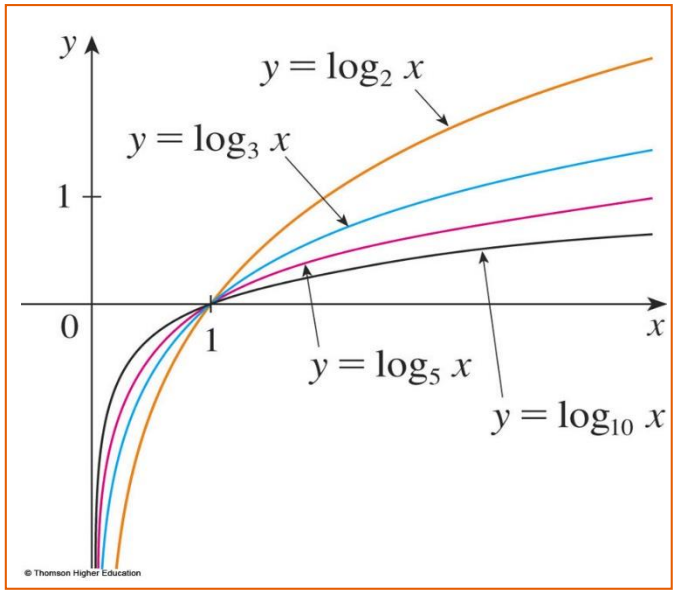
TRANSCENDENTAL FUNCTIONS

LOGARITHMIC FUNCTIONS

The logarithmic functions $f(x) = \log_a x$

where the base *a* is a positive constant, are the inverse functions of the exponential functions.

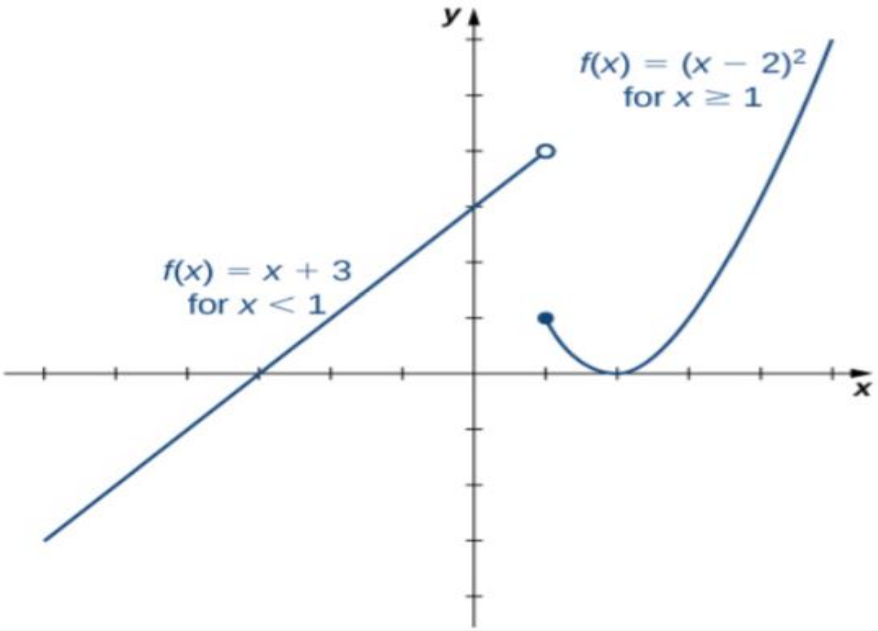
The figure shows the graphs of four logarithmic functions with various bases.



PIECEWISE-DEFINED FUNCTIONS

Example:

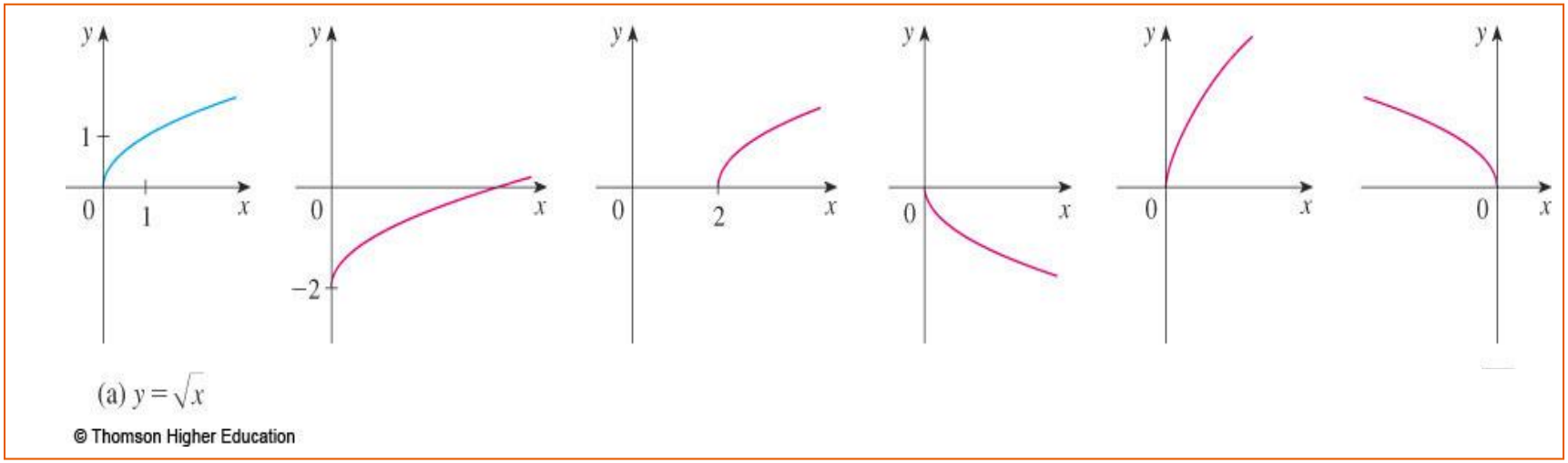
$$f(x) = \begin{cases} x+3, & x < 1 \\ (x-2)^2, & x \geq 1 \end{cases}.$$



TRANSFORMATIONS OF FUNCTION

Label the following graph from the graph of the function y=f(x) shown in the part (a)

y=f(x)-2, y=f(x-2), y=-f(x), y=2f(x), y=f(-x)?



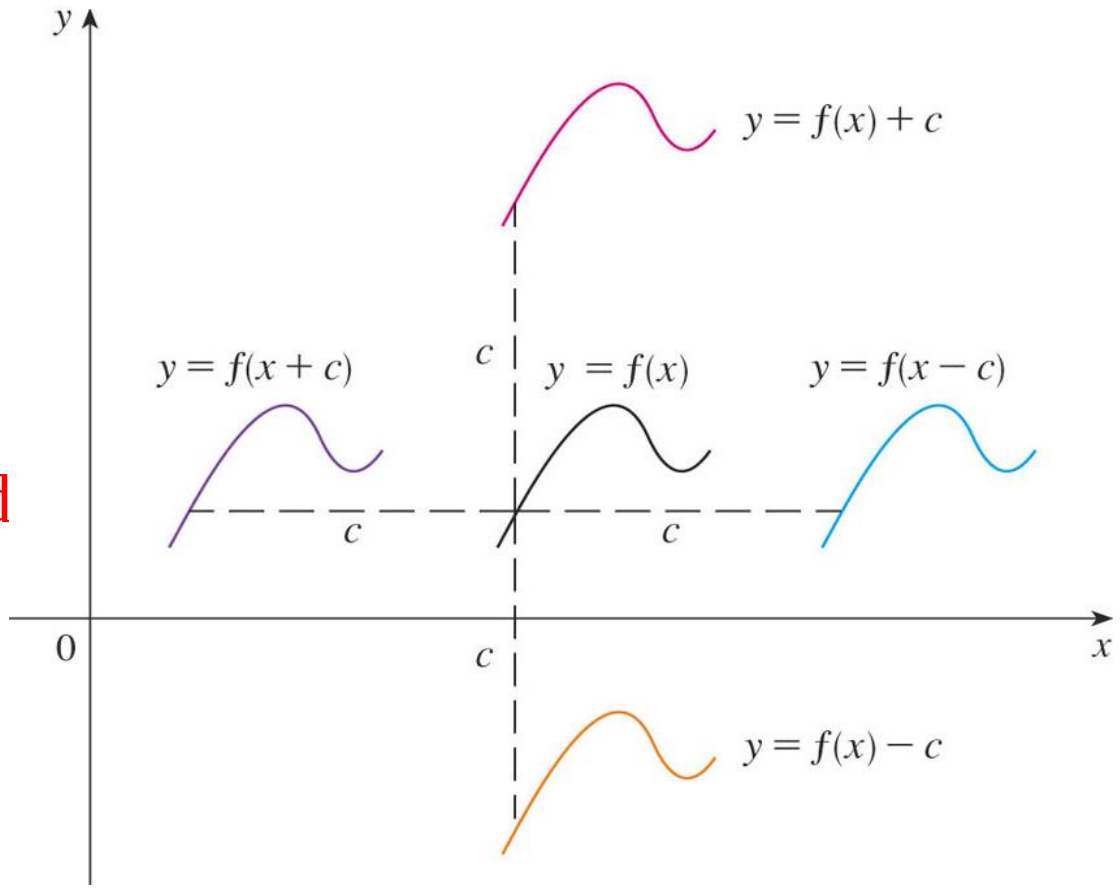
(a) $y = \sqrt{x}$

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SHIFTING

Why don't we consider the case c<0?

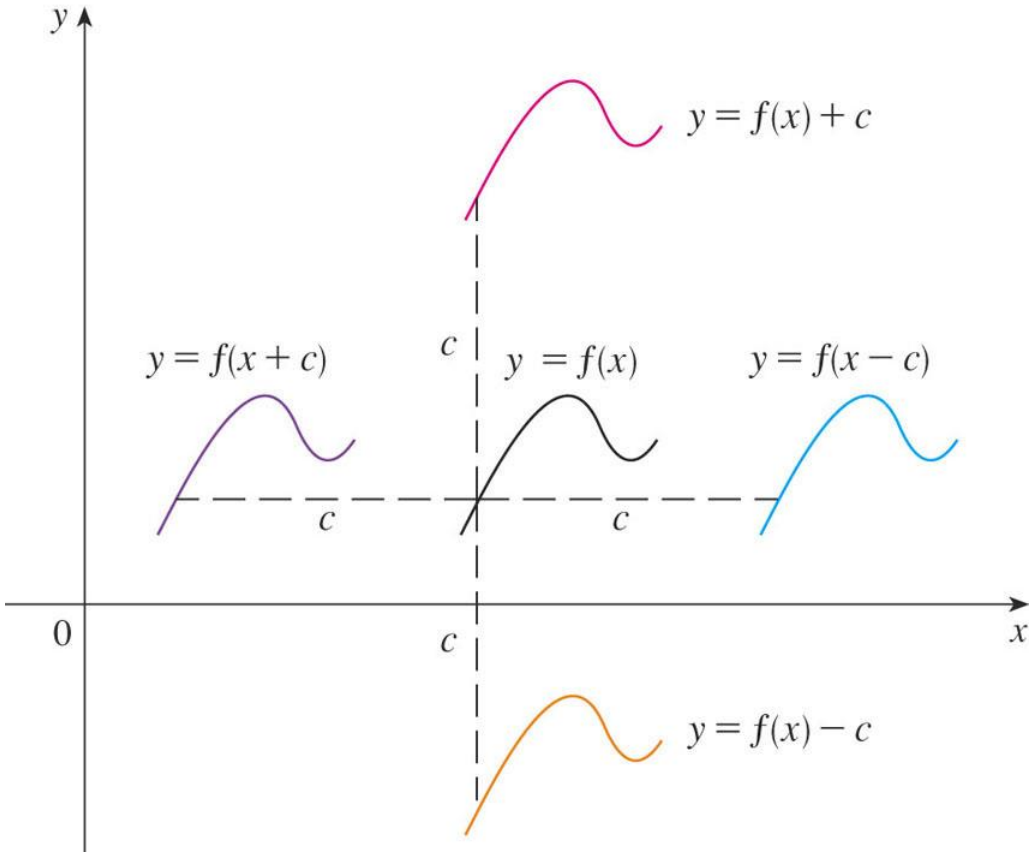
- Suppose $c > 0$.
  - To obtain the graph of $y = f(x) + c$, shift the graph of $y = f(x)$ a distance c units upward
  - To obtain the graph of $y = f(x) - c$, shift the graph of $y = f(x)$ a distance c units downward.



SHIFTING

- To obtain the graph of $y = f(x - c)$, shift the graph of $y = f(x)$ a distance c units to the right.

- To obtain the graph of $y = f(x + c)$, shift the graph of $y = f(x)$ a distance c units to the left.



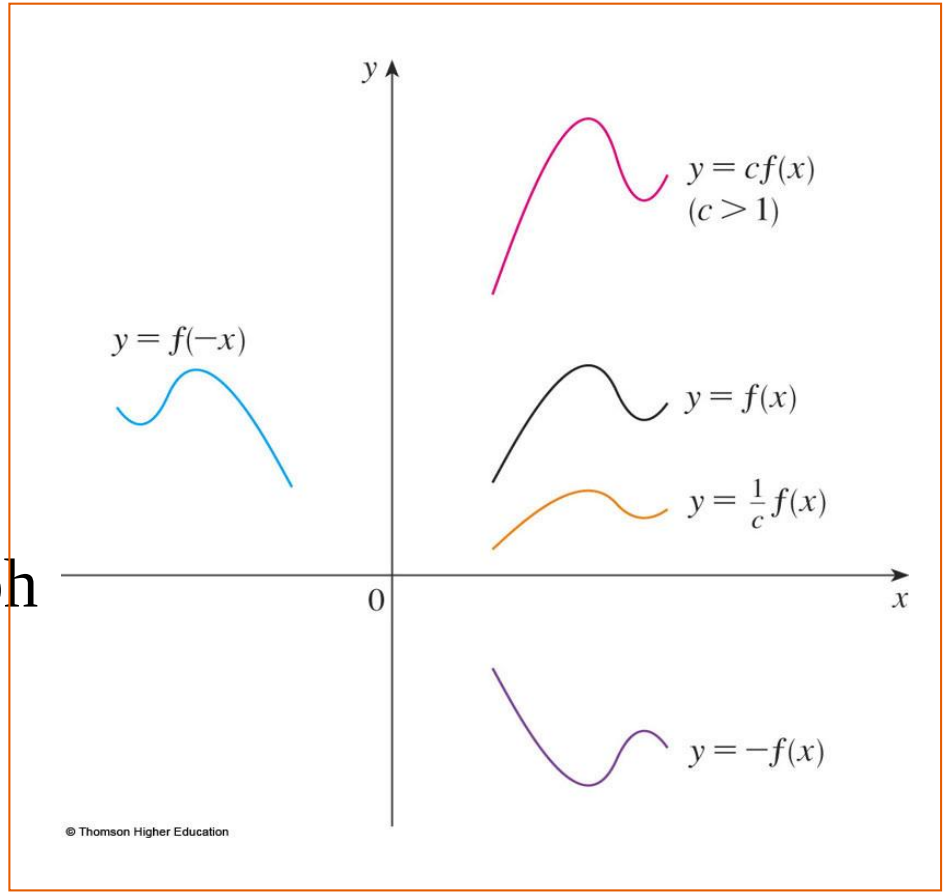
TRANSFORMATIONS

How about the case c<1?

- Suppose $c > 1$.

- To obtain the gr of $y = c\,f(x)$, stretch the graph of $y = f(x)$ vertically by a factor of c.

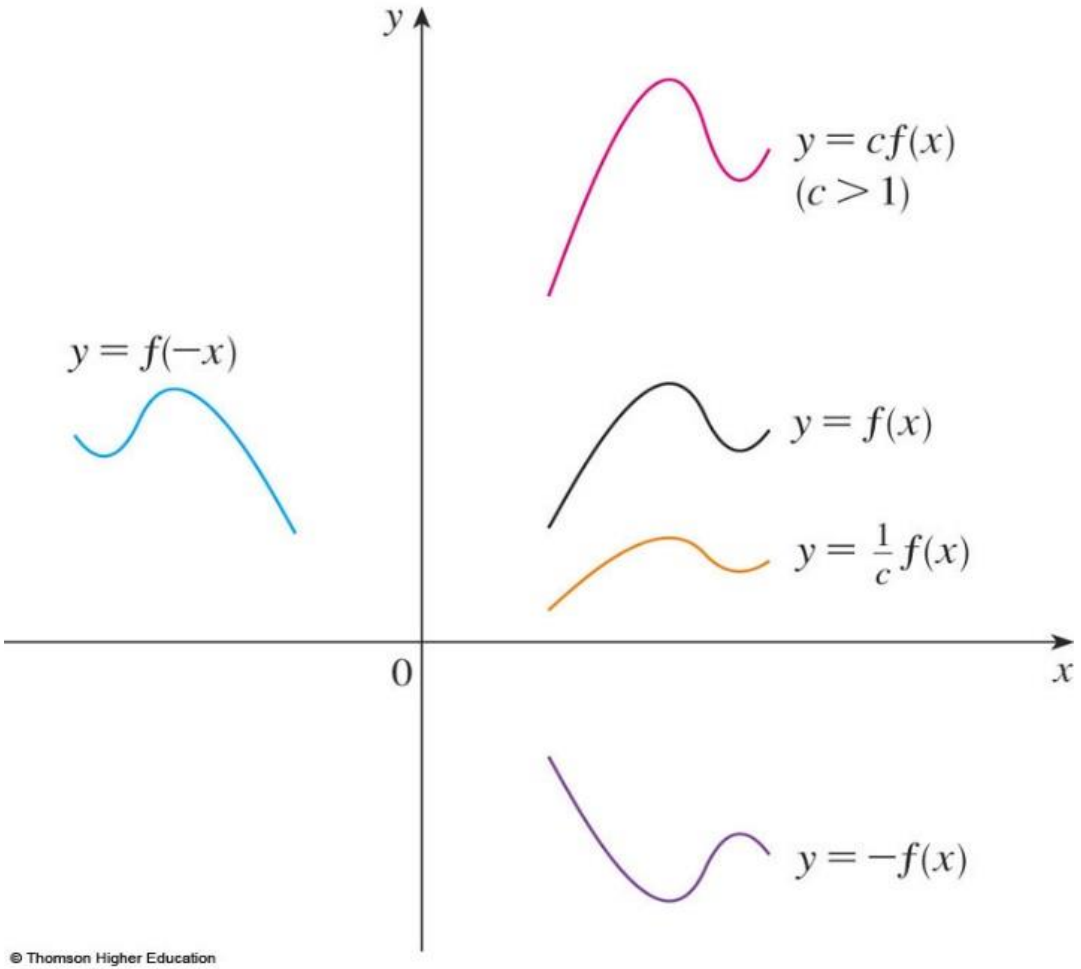
- To obtain the graph of $y = (1/c)f(x)$, compress the graph of $y = f(x)$ vertically by a factor of c.



TRANSFORMATIONS

– To obtain the graph of $y = f(cx)$, compress the graph of $y = f(x)$ horizontally by a factor of c.

– To obtain the graph of $y = f(x/c)$, stretch the graph of $y = f(x)$ horizontally by a factor of c.

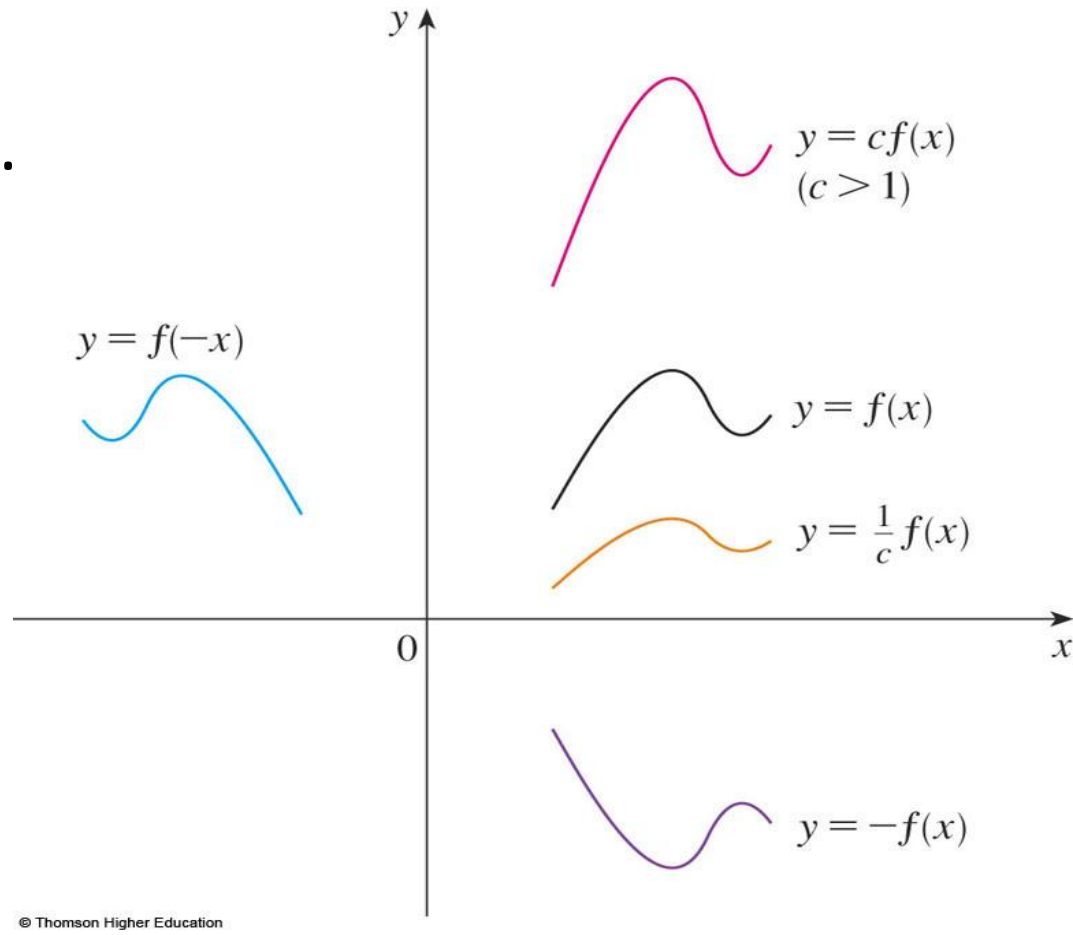


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TRANSFORMATIONS

– To obtain the graph of $y = -f(x)$, reflect the graph of $y = f(x)$ about the x-axis.

– To obtain the graph of $y = f(-x)$, reflect the graph of $y = f(x)$ about the y-axis.



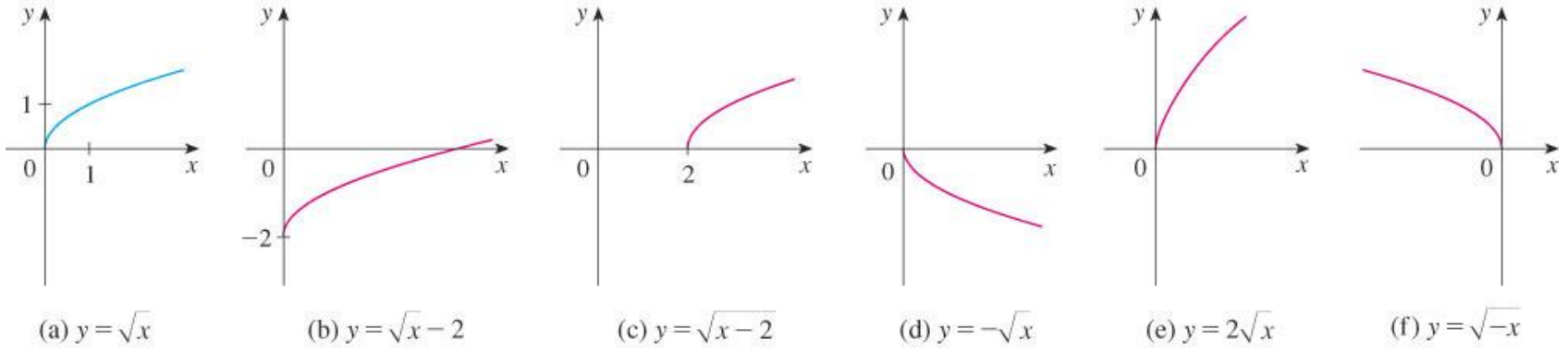
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NEW FUNCTIONS FROM OLD FUNCTIONS

Label the following graph from the graph of the function $y = \sqrt{x}$ shown in the part (a):

y=f(x)-2, y=f(x-2), y=-f(x), y=2f(x), y=f(-x)?

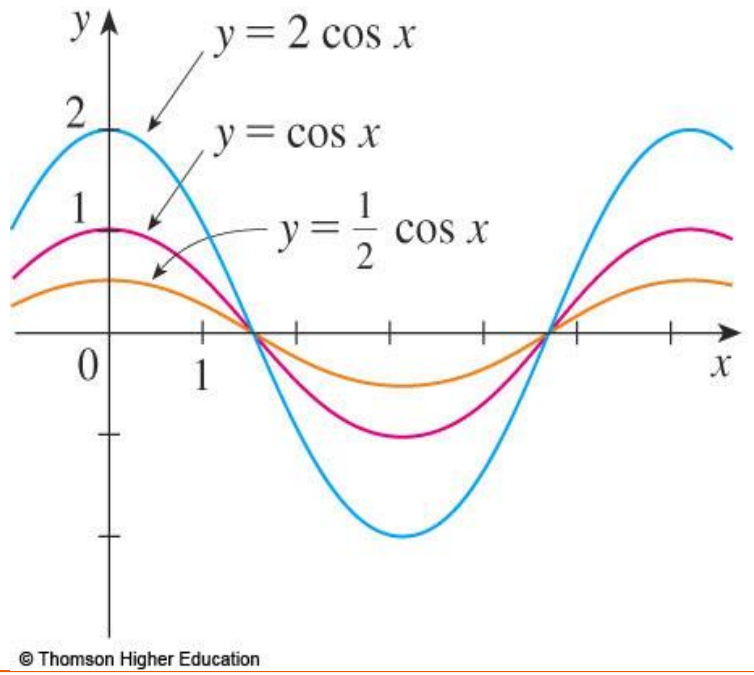
- $y = -\sqrt{x}$ by reflecting about the *x*-axis.
- $y = 2\sqrt{x}$ by stretching vertically by a factor of 2.
- $y = \sqrt{-x}$ by reflecting about the *y*-axis

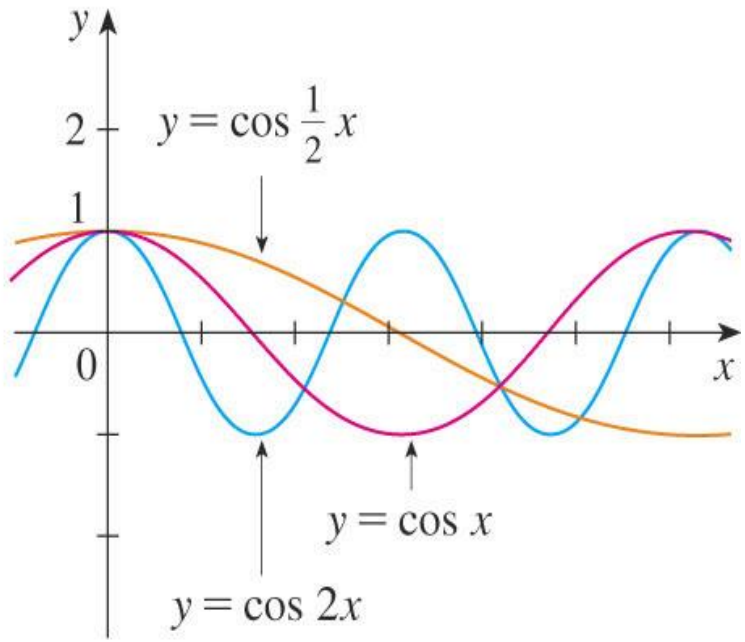


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TRANSFORMATIONS

- The figure illustrates these stretching transformations when applied to the cosine function with $c = 2$.





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Example

Suppose that the graph of f is given.
Describe how the graph of the function $f(x-2)+2$ can be obtained from the graph of f.
Select the correct answer.

a. Shift the graph 2 units to the left and 2 units down.

b. Shift the graph 2 units to the right and 2 units down.

c. Shift the graph 2 units to the right and 2 units up.

d. Shift the graph 2 units to the left and 2 units up.

e. none of these

QUIZ QUESTIONS

Label the following graphs: $f(x)$, $\frac{1}{2}f(x)$, $f\left(\frac{1}{2}x\right)$.

